



BACHELOR OF INFORMATION TECHNOLOGY (HONS)

**FINAL EXAMINATION
JULY 2022**

Course : EC3123/EC3239 (Operating Systems Principles) Time: 2.00pm – 5.00pm
(3 hours)

Lecturer: Wan Nor Al-Ashekin Bt. Wan Husin

Date: 11 January 2023

Instructions:

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This question paper consists of **3** pages.
(excluding front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

1. a. What is operating system? (6 MARKS)
- b. Discuss **THREE (3)** aims of operating system in providing supports for the system components. (6 MARKS)
- c. Describe any **TWO (2)** tasks that the subsystem manager must perform in operating system. (4 MARKS)
- d. Network manager provides a convenient way for users to share resources while controlling users' access to it. (4 MARKS)
Explain with examples **TWO (2)** categories of network manager resources.
2. a. Explain **TWO (2)** communication models in operating system. (6 MARKS)
- b. Table 1 contains the process ID, arrival time, CPU time and priority:

Table 1

Process ID	Arrival Time	CPU Time	Priority
P1	0	20	3
P2	1	15	2
P3	2	25	1
P4	3	10	4
P5	4	5	5

Based on the information in Table 1, draw the Gantt Chart for the following process scheduling:

- i. Priority Scheduling. (1 MARK)
- ii. Round Robin Scheduling with Time Quantum = 20 ms. (2 MARKS)
- iii. Based on the information in Table 1 and Gantt Chart in Question 2b(i) and 2b(ii), write the Waiting Time for Priority Scheduling and Round Robin Scheduling. (5 MARKS)
- c. Explain any **THREE (3)** conditions of deadlocks in operating system. (6 MARKS)

3. a. Describe the concept of swapping in memory. (2 MARKS)
- b. Explain **TWO (2)** types of fragmentations in memory with example of figure. (8 MARKS)
- c. Discuss **TWO (2)** problems in memory paging and **TWO (2)** solutions for each problem. (10 MARKS)
4. a. Explain the following types of file access in operating system: (8 MARKS)
i. Sequential File
ii. Indexed Sequential File
- b. Describe with an example, the character-oriented device and block-oriented device. (6 MARKS)
- c. Discuss **SIX (6)** typical duties of a clock driver in in device management. (6 MARKS)
5. a. List any **FOUR (4)** system files in Ubuntu Linux operating system. (4 MARKS)
- b. Assume that the current working directory is /home/ricky. Write an appropriate Linux commands for the following statements: (10 MARKS)
i. Ricky wants to create two empty directories called *EC3123_Exercise* and *EC3123_Assignment*.
ii. Ricky changes to directory *EC3123_Exercise* and wants to create two files called *Exercise1* and *Exercise 2*.
iii. Ricky wants to remove file *Exercise 1* from the system.
iv. Ricky wants to remove directory *EC3123_Exercise* and any files and subdirectories it contains.
v. Ricky wants to view the list of files and file permissions in directory *EC3123_Assignment*.
vi. Ricky wants to add a new user with user's home directory called *Yusma* in the system.

- vii. Ricky wants to set a password for user *Yusma*.
 - viii. Ricky wants to add a new group called *SeniorGroup* with primary group ID *5100*.
 - ix. Ricky wants to create a text file called *Reports.txt* and insert some texts in the file.
 - x. Ricky wants to view the list of files in directory */etc* from home directory.
- c. Write a single Ubuntu Linux command in octal form to set the following permissions for the file name *assignment*. **(6 MARKS)**
- i. `rw-rw-rwx :`
 - ii. `r-xr-xr-x :`
 - iii. `r--r--r-- :`
 - iv. `rwX--- :`
 - v. `rw-r--- :`
 - vi. `rw-rw-rw- :`

-END OF QUESTION PAPER-